Instructor notes: same evolution, different probabilities

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Supplementary material

1 Use and prerequisites

These are proposed activities, not a report of a teaching intervention. The core work needs normalised complex vectors, the Born rule, a geometric series and Hermitian 2×2 eigenvalues. The multiple-outcome extension is suitable for a computational project after the two-mode lesson. A possible allocation is one tutorial for predictions and derivations, followed by a separate computing session. Instructors should adapt this to students' background rather than treat it as a tested schedule.

2 Worked solutions

2.1 Predicting from the angle error

The angle errors alone do not determine the ranking: the probability comparison includes two different readout effects. An estimate comparing two evolutions followed by the *same* measurement cannot be applied without a further readout term. The five-node grid has a node at $1/2$; the six-node grid has equal populations on both halves. Ask students to predict the constant-state outcome before introducing superpositions.

2.2 Continuous measurement

The diagonal integral is $1/2$. The upper off-diagonal integral is $\int_0^{1/2} e^{2\pi \mathrm{i}\theta} d\theta = \mathrm{i}/\pi$. Therefore

$$E = \begin{pmatrix} 1/2 & \mathrm{i}/\pi \\ -\mathrm{i}/\pi & 1/2 \end{pmatrix}, \quad \lambda_\pm = 1/2 \pm 1/\pi.$$

The eigenvalues are between zero and one, so $E, I - E$ form a valid measurement, but they are not idempotent. For the full coordinate projector P_0 and the two-mode projector Π, the effect is $\Pi P_0 \Pi$ restricted to the input subspace. Generally $(\Pi P_0 \Pi)^2 \neq \Pi P_0 \Pi$: multiplication by the half-circle indicator creates modes outside the input space. This also explains why the compressed POVM does not specify repeated coordinate-measurement back-action. The balanced state gives $p_0 = 1/2 - \sin\phi/\pi$; the sign is a useful check.

2.3 Node sums

If there are L nodes in the first half,

$$g_Q = \frac{1}{Q}\sum_{j=0}^{L-1} e^{2\pi \mathrm{i}(j+1/2)/Q} = \frac{e^{\mathrm{i}\pi L/Q}\sin(\pi L/Q)}{Q\sin(\pi/Q)}.$$

For even Q, $L = Q/2$; for odd Q, $L = (Q-1)/2$. Substitution gives the parity formulas in the paper. In the odd case the real part of g_Q is $1/(2Q)$, while its imaginary part is $\cot(\pi/(2Q))/(2Q)$. The constant input has error $1/(2Q)$, which is only a lower bound on the worst-case error over superpositions.

2.4 Power-of-two registers

For nodes j/Q with even Q, the first half contains $j = 0, \ldots, Q/2-1$, so $d_Q = 0$. Using $1/(1-e^{\mathrm{i}x}) = \frac{1}{2} + \frac{\mathrm{i}}{2}\cot(x/2)$,

$$g_Q = \frac{1}{Q}\sum_{j=0}^{Q/2-1} e^{2\pi \mathrm{i} j/Q} = \frac{2}{Q\,(1-e^{2\pi \mathrm{i}/Q})} = \frac{1+\mathrm{i}\cot(\pi/Q)}{Q}.$$

Since $\cot(\pi/Q)/Q = 1/\pi - \pi/(3Q^2) + O(Q^{-4})$, the off-diagonal error at $k=0$ is dominated by its real part $1/Q$. Ask students why the constant wavefunction cannot detect this error, whereas a balanced superposition can. For $Q=8$, $a=5$ the errors at $k=(0,1)$ are $(0.126089, 0.112252)$ for $\xi = 0$ and $(0.008331, 0.016388)$ for $\xi = 1/2$.

An exercise on the boundary convention: keep the nodes j/Q but assign each of the two boundary nodes to either outcome with probability $1/2$. Their diagonal contributions still total $1/Q$, so $d_Q = 0$, while their off-diagonal contributions $1/(2Q)$ and $-1/(2Q)$ cancel. Hence $g_Q = \mathrm{i}\cot(\pi/Q)/Q$ and $\varepsilon_0 = 1/\pi - \cot(\pi/Q)/Q \approx \pi/(3Q^2)$, which is 0.0165 at $Q=8$. Ask students why this is a change of the effect, not of the evolution.

2.5 Worst input and exact error

Let $h = |h|e^{\mathrm{i}\gamma}$ and $\Delta = \left(\begin{smallmatrix} d & h \\ h^* & d \end{smallmatrix}\right)$. The equal-weight vectors

$$c_+ = \frac{(1, e^{-\mathrm{i}\gamma})^T}{\sqrt{2}}, \qquad c_- = \frac{(1, -e^{-\mathrm{i}\gamma})^T}{\sqrt{2}}$$

give expectation values $d + |h|$ and $d - |h|$. Use c_- for odd midpoint grids, where $d < 0$; either extremum works when $d = 0$. If $h = 0$ every state attains $|d|$. These vectors are in the frame with the ideal evolution removed; the physical input for the original comparison is $D_\rho(k)^* c_\pm$. Thus $\varepsilon_k = |d| + |h|$. Students should distinguish this from keeping one maximising state fixed as k varies.

For $\rho = (\sqrt{5}-1)/2$ at $k=1$:

	one midpoint grid	two shifted grids
$Q=5$, $a=3$	0.236256	0.036731
$Q=6$, $a=4$	0.100279	0.097513

The five-node approximation is closer in angle but initially worse in probability. Applying the same two-setting remedy to both reverses this ranking.

2.6 Shifts, dynamics and resources

For $r = 0, 1$, the nodes $(j + (r + 1/2)/2)/Q$ are precisely the midpoint nodes of $2Q$ when interlaced. Averaging therefore reproduces each matrix element of the $2Q$-node effect. Each branch still implements angle a/Q. The mixture changes a measurement, not the angle generator. A random bit selects the encoding branch; it is not necessary to create both copies of a state in one run. Extra encoding control and sampling are nevertheless required. A fixed unshifted encoding cannot simply be relabelled to achieve the same effect, because off-diagonal matrix elements also depend on the shift.

2.7 Controlled comparison and assessment rubric

Ask students to predict whether 3/5 on five nodes and 6/10 on ten nodes must have equal errors. The propagators on the two-mode space are identical, but the detector effects are not. The fraction 6/10 is deliberately not reduced: it specifies a permutation on a ten-node grid. The errors are respectively $(0.200554, 0.236256)$ and $(0.005297, 0.036731)$ at $k = (0, 1)$. Next compare 3/5 with 4/6: the former has smaller angle error but larger probability error at $k = 1$. Finally identify the ten-node effect with the average of the two five-node settings. This equality concerns terminal probabilities, not implementation cost.

Award one point for each of: (i) identifying the fixed propagator in the first comparison; (ii) correctly calculating the two readout effects; (iii) obtaining the one-step ranking for 3/5 versus 4/6; (iv) explaining that a bound for one common measurement cannot account for a change in the detector effect. A response mentioning only parity earns no point for (iv). This rubric is proposed, not empirically validated.

A useful check on overinterpretation is 5/8 versus 6/10, both on even midpoint grids. Their readout errors at $k = 0$ are 0.008331 and 0.005297, but their one-step errors are 0.016388 and 0.036731. This is not another one-step reversal: at $k = 0$ the evolution has not occurred.

3 Computational project

Run `teaching.py` from the code repository https://github.com/polushkin-ayu/two-mode-readout with Python, NumPy, SciPy and Matplotlib. The script writes `two-mode-results.json`, `parameter-sweep.csv` to `results/` and the figure to `results/figures/`. It chooses the nearest numerator $a = \mathrm{round}(Q\rho)$ and evaluates all integer times through K. There are 144 parameter combinations: three rotation angles, eight node counts, two shift counts and three horizons. These finite comparisons do not establish global optimality.

Suggested student outputs are: a reproduction of table 1 of the paper, a log-log plot of readout error at $k = 0$ separating odd and even grids, and one explanation of a change in ranking with horizon. For the log-log plot use a longer sequence, for example $Q = 5, \ldots, 200$, rather than fit an exponent to two points. Keep the two-mode input space fixed.

Expected asymptotics are $\varepsilon_0 \sim 1/Q$ for odd midpoint grids and $\varepsilon_0 \sim \pi/(6Q^2)$ for even grids. Two shifts give $\varepsilon_0 \sim \pi/(24Q^2)$ for either parity. At later times a small angle difference gives a contribution on the scale $k|\delta|$ until periodicity matters. Neither

long-time monotonic growth nor one permanent ranking should be inferred from that small-error approximation.

A useful extension is to compare a prescribed initial state with the worst-case curve, explicitly reporting the state used. Another is to repeat a binary measurement halfway through the evolution: this requires a measurement instrument and is *not* determined by the terminal POVM alone. It should be posed as a question about missing modelling assumptions, not as a substitution into the same formula.

4 Readout convergence summary

All statements below concern $k = 0$, fixed input modes and fixed detector sectors. For the two-mode binary problem,

midpoint, odd Q	$\varepsilon_0 \sim Q^{-1}$
midpoint, even Q	$\varepsilon_0 \sim \pi/(6Q^2)$
two shifts, either parity	$\varepsilon_0 \sim \pi/(24Q^2)$

For the graduate extension with fixed M, B, $B \mid Q$ gives worst-case readout error $O(Q^{-2})$. If $Q = BL + r$, $0 < r < B$, the constant mode has exact TV error $r(B - r)/(BQ)$; this is a lower bound, not an exact formula for the worst-case error over all modes. Increasing M requires new estimates.

5 Questions for formative assessment

1. Can an exact propagator give an inaccurate predicted detector probability? Explain with a fixed finite node readout.

2. Why does the constant wavefunction already detect an odd-grid bias, but fail to attain the full worst-case error?

3. Why is a positive matrix with eigenvalues in $[0, 1]$ an admissible effect even when it is not a projector?

4. What is held fixed when taking the limit $Q \to \infty$ in the error orders? Why would increasing the mode cutoff require a new estimate?

5. Does equality of terminal probabilities imply equal states after measurement? Which additional objects would be needed to answer that?

Correct answers should separate dynamics, input class, detector effects and instruments. The intended learning outcome is this separation, rather than memorising which particular denominator won the example.